

PART-B

MODEL ANSWERS OF END-SEMESTER EXAMINATION (TOTAL POINTS: 20)

1. (6 points) Let

$$\mathbf{S} = \begin{pmatrix} S_{yy} & S_{y\mathbf{X}} \\ S_{y\mathbf{X}} & S_{\mathbf{X}\mathbf{X}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $\mathbf{X}_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\mathbf{X}$.

Solution: In standard CCA, the squared sample canonical correlations r^2 are the eigenvalues of the matrix

$$\mathbf{S}_{11}^{-\frac{1}{2}} \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21} \mathbf{S}_{11}^{-\frac{1}{2}} = \frac{1}{s_y^2} \mathbf{s}_{y\mathbf{X}}^T \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y}.$$

Notice that $\mathbf{s}_{y\mathbf{X}}^T$ is a $1 \times q$ vector, $\mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1}$ is a $q \times q$ matrix, and $\mathbf{s}_{\mathbf{X}y}$ is a $q \times 1$ vector. The resulting product is a 1×1 scalar. The only eigenvalue of a scalar is the scalar itself. Therefore, the single squared canonical correlation is:

$$r_1^2 = \frac{\mathbf{s}_{y\mathbf{X}}^T \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y}}{s_y^2}. \quad (1)$$

Now, consider the multiple linear regression of Y on the variables in $\mathbf{X}$. The estimated regression coefficient vector is given by the standard ordinary least squares (OLS) solution:

$$\hat{\boldsymbol{\beta}} = \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y}$$

The portion of the variance in Y explained by the regression model (the Explained Variance) is calculated as $\hat{\boldsymbol{\beta}}^T \mathbf{s}_{\mathbf{X}y}$:

$$SS_{Reg} = (\mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y})^T \mathbf{s}_{\mathbf{X}y} = \mathbf{s}_{y\mathbf{X}}^T (\mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1})^T \mathbf{s}_{\mathbf{X}y} = \mathbf{s}_{y\mathbf{X}}^T \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y},$$

since $(\mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1})^T = \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1}$. Therefore,

$$R^2 = \frac{SS_{Reg}}{SS_T} = \frac{\mathbf{s}_{y\mathbf{X}}^T \mathbf{S}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{s}_{\mathbf{X}y}}{s_y^2}. \quad (2)$$

By comparing equations (1) and (2), we have $r_1^2 = R^2$.

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.

Solution: In the principal component method of factor analysis, the columns of the estimated factor loading matrix $\tilde{\mathbf{L}}$ are derived as follows: First, the eigenvalues are arranged in descending order of magnitude. Then, the largest m eigenvalues and corresponding normalized eigenvectors by the square root of their corresponding eigenvalues.

Since $m = 2$, the 4×2 matrix $\tilde{\mathbf{L}}$ is given by:

$$\tilde{\mathbf{L}} = [\sqrt{\lambda_3}\mathbf{e}_3 \quad \sqrt{\lambda_2}\mathbf{e}_2].$$

- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.

Solution: The estimated communality for the i -th variable, $\tilde{h}_i^2$, represents the portion of the variable's variance explained by the m common factors. It is calculated as the sum of squared loadings for that variable (i.e., the squared Euclidean norm of the i -th row of $\tilde{\mathbf{L}}$).

$$\tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2 \implies \tilde{h}_1^2 = (0.8)^2 + (-0.4)^2 = 0.80.$$

- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

Solution: Let $\tilde{l}_{ij}$ be the element in the i -th row and j -th column of $\tilde{\mathbf{L}}$. The sum of all estimated communalities is:

$$\sum_{i=1}^p \tilde{h}_i^2 = \sum_{i=1}^p \left(\sum_{j=1}^m \tilde{l}_{ij}^2 \right).$$

By interchanging the order of summation, we get:

$$\sum_{i=1}^p \tilde{h}_i^2 = \sum_{j=1}^m \left(\sum_{i=1}^p \tilde{l}_{ij}^2 \right).$$

The inner sum $\sum_{i=1}^p \tilde{l}_{ij}^2$ is the sum of squared elements in the j -th column of $\tilde{\mathbf{L}}$. From Part (a), the j -th column is $\sqrt{\lambda_j}\mathbf{e}_j$. The squared norm of this column is:

$$(\sqrt{\lambda_j}\mathbf{e}_j)^T(\sqrt{\lambda_j}\mathbf{e}_j) = \lambda_j(\mathbf{e}_j^T\mathbf{e}_j).$$

Because the eigenvectors are normalized (orthonormal), $\mathbf{e}_j^T\mathbf{e}_j = 1$. Therefore, the inner sum equals λ_j . Substituting this back yields:

$$\sum_{i=1}^p \tilde{h}_i^2 = \sum_{j=1}^m \lambda_j.$$

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Solution: Using CLT, we have

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} Z_1 \sim N(0, c\mu^2).$$

Using delta method,

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} Z_2 \sim N(0, c\mu^2(g'(\mu))^2).$$

Thus, we want $g(\cdot)$ such that

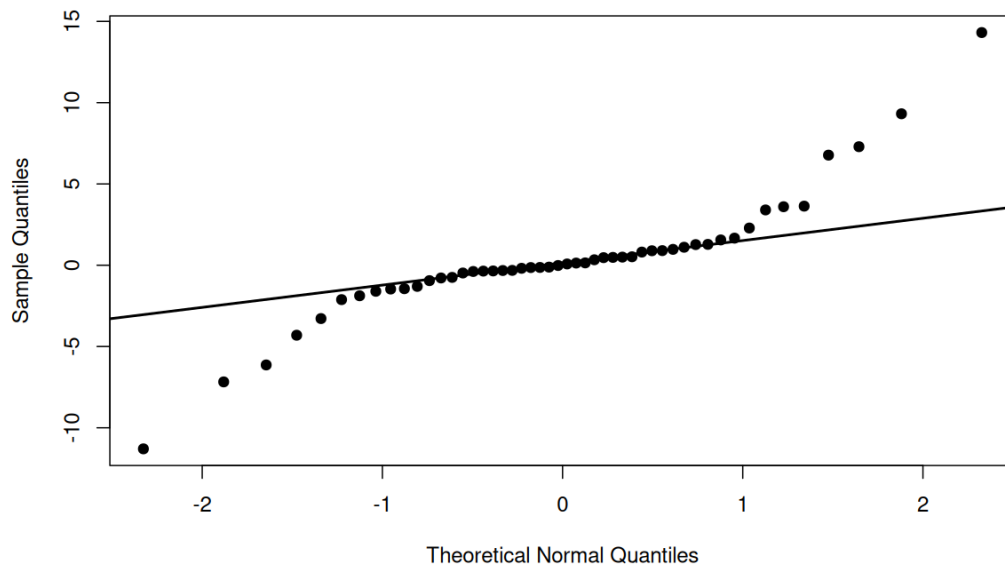
$$\mu^2(g'(\mu))^2 = k \quad (\text{some constant } k).$$

Therefore, a variance-stabilizing transformation satisfies

$$g'(\mu) = \frac{1}{\mu} \implies g(\mu) = \ln \mu.$$

Thus, $g(x) = \ln x$ is a variance-stabilizing transformation.

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Solution: The S-shaped curvature indicates that the distribution has heavier tails than a normal distribution. The data is "fat-tailed," meaning extreme outliers are more frequent than the normal model predicts. Therefore, a heavy tailed distribution (like Cauchy distribution or t -distribution with low degrees of freedom) would provide better fit to the model.